

5. Traceability Matrix

Requirement ID	Requirement Description	Testing Technique	Test Cases / Section	Result
R1	The same input should always generate identical serialized output under the same Python version	Determinism Testing, Equivalence Partitioning	Section 2.1 (34 determinism test cases)	Pass
R2	Primitive data types should be serialized and deserialized correctly	Equivalence Partitioning	None, bool, int, float, string, bytes, complex	Pass
R3	Container types should be serialized and deserialized correctly	Equivalence Partitioning	List, Tuple, Dict, Set, FrozenSet, Nested Containers	Pass
R4	Floating-point special values should be handled correctly	Equivalence Partitioning, Boundary Value Analysis	NaN, +Inf, -Inf, -0.0, FloatEpsilon, FloatMax	Pass
R5	Recursive and cyclic data structures should be handled correctly	Special Case Testing	SelfReferentialList, SelfReferentialDict, MutualReference, TripleCycle, CycleInTuple, CycleInDict	Pass
R6	Deeply nested structures should be detected safely	White-box Testing	DeepNested (2000 levels)	Pass (Expected Exception)
R7	Integer boundary values should be handled correctly	Boundary Value Analysis	$2^{31}-1$, -2^{31} , 2^{31} , $2^{63}-1$, -2^{63} , 2^{63} , 2^{1000}	Pass
R8	String length boundary values should be handled correctly	Boundary Value Analysis	Length 0, 1, 255, 256, 65535, 65536	Pass

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R9	Empty and large collections should be handled correctly	Boundary Value Analysis	Empty/List/Tuple/Dict, Large collections (10,000 elements)	Pass
R10	Serialization and deserialization should preserve object values	Round-trip Verification	All determinism, boundary, recursion, and fuzzing tests	Pass
R11	Randomly generated valid objects should be handled correctly	Fuzz Testing	100 generated random objects	Pass
R12	Major serialization paths in marshal.c should be exercised	White-box Testing	TYPE_NONE, TYPE_BOOL, TYPE_INT, TYPE_FLOAT, TYPE_COMPLEX, TYPE_STRING, TYPE_UNICODE, TYPE_TUPLE, TYPE_LIST, TYPE_DICT, TYPE_SET, TYPE_FROZENSET	Pass
R13	Reference tracking mechanism should be exercised	White-box Testing	TypeRef	Failed (System Recursion Limit)
R14	Different Python versions may produce different serialized outputs	Differential Testing	Python 3.9, 3.11, 3.12, 3.13	Differences Observed
R15	Different operating systems should not affect basic object serialization	Cross-platform Testing	Windows 10 and macOS	Pass for basic types
R16	Unsupported objects should raise appropriate exceptions	Lightweight Testing	Slice Object	Pass

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R17	Special marshal-supported objects should be serialized correctly	Lightweight Testing	Ellipsis, StopIteration	Pass
R18	Code objects should be examined for version compatibility	Lightweight Testing	code_object, pyc_timestamp_mode, pyc_hash_mode	Incompatible Across Versions